Engineering Physics

Unit 5: Crystal Structure and X-Rays

Unit cell, Bravais lattice, cubic system, number of atoms per unit cell, coordination number, atomic radius, packing density, relation between lattice constant and density, lattice planes and Miller indices, Interplaner spacing for cubic system, Bragg's law, X-ray diffraction, Applications of X-ray

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Solid State: In the solid state, atoms or molecules are tightly packed and have relatively little kinetic energy. They vibrate in fixed positions but do not have enough energy to break away from their neighbors. As a result, solids maintain a definite shape and volume. However, the arrangement of atoms in solids can vary, leading to two main categories: amorphous and crystalline solids.

Amorphous Solids: In amorphous solids, the arrangement of atoms or molecules lacks a regular, repeating pattern. These solids have closely packed atoms, but the positions are random and disordered. Common examples of amorphous solids include glass, rubber, and some plastics. Because of their disordered structure, amorphous solids often have properties such as transparency, lack of a distinct melting point, and the ability to be molded into various shapes.

Crystalline Solids: In crystalline solids, the atoms or molecules are arranged in a highly ordered and repeating three-dimensional pattern called a crystal lattice. This regular arrangement imparts distinct characteristics to crystalline solids, such as a well-defined melting point, cleavage planes, and the ability to form well-defined, geometrically shaped crystals. Common examples of crystalline solids include salt, diamond, and most metals.

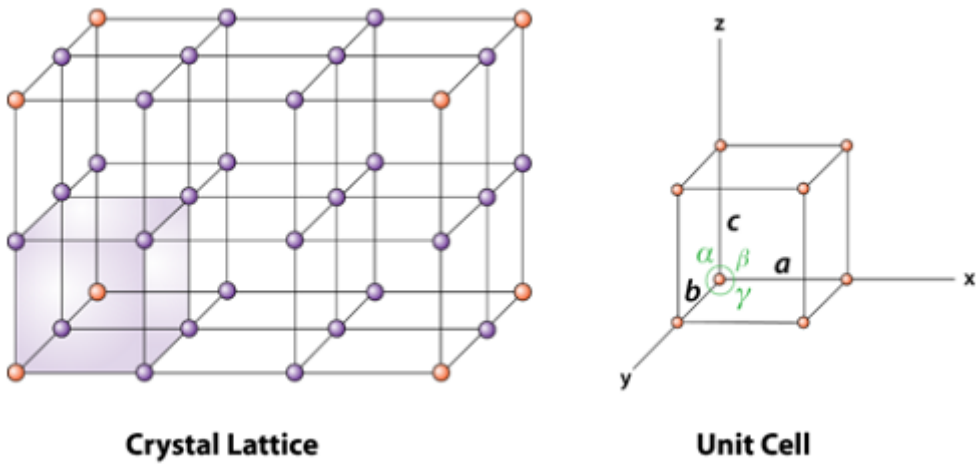
Lattice Points:

Lattice points are imaginary points within a crystal that represent the positions of the repeating structural units, whether they are atoms, ions, or molecules. Each lattice point represents the center of one such structural unit. Lattice points do not have any physical size, and they are used to describe the spatial arrangement and symmetry of the crystal's atomic or molecular structure. The repeating pattern of lattice points is responsible for the overall crystalline symmetry.

Space Lattice:

A space lattice, also known as a Bravais lattice, is a 3D lattice structure that extends in three dimensions. It is the most complex and versatile of the three lattice types and serves as the foundation for describing the arrangement of atoms or molecules in crystalline materials.

A space lattice is defined as an array of point (collection of points) arranged in a regular periodic manner in three dimensional space such that each point in the lattice has exactly identical surrounding.



Basis: The atom or group of atoms that is associated with every lattice point is called basis.

Group of atoms or molecules identical in composition.

Lattice + basis = crystal structure

In summary, lattice points are imaginary points representing the positions of repeating structural units within a crystal. Linear lattices are the simplest 1D representations of these points, while space lattices extend to 3D and provide a framework for describing the arrangement of atoms or molecules in crystalline materials with varying degrees of symmetry and complexity.

A unit cell of space lattice is defined as the smallest closed portion of the lattice such that the lattice is form by the repetition of smallest closed portion.

OR

The basic or fundamental building block of volume by which the crystal can be constructed by three dimensional translational repetition of it.

Or

Unit cell is

•small repeat entity,

•basic structural unit

•building block of crystal structure

Lattice parameters -The size and shape of the unit cell can be described by **three lattice vectors**: a, b, c or the axial lengths a, b, and c and **interaxial angles** α, β, γ

Primitive Unit Cell- In primitive Unit cell atoms are located at the corners only.

Ex. Simple Cubic Structure (SC)

Non –Primitive unit cell-- In non-primitive Unit cell atoms are located not only at the corners but also inside the body.

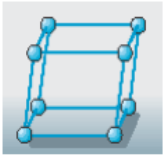
Ex. BCC, FCC

Bravais Lattice

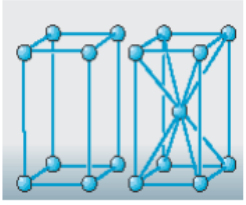
The unit vectors a, b and c are called lattice parameters. Based on their length equality or inequality and their orientation (the angles between them, α, β and γ) a total of 7 crystal systems can be defined. With the centering (face, base and body centering) added to these, 14 kinds of 3D lattices, known as Bravais lattices, can be generated.

S. No	Name of the crystal systems	Primitives	Interfacial angles	Bravais Lattice points
1	Cubic	a= b= c	α=β=γ=90°	3(P,I,F)
2	Tetragonal	a= b≠ c	α=β=γ=90°	2(P,I)
3	Orthorhombic	a≠ b≠ c	α=β=γ=90°	4(P,C,I,F)
4	Trigonal (Rhombohedral)	a= b= c	α=β=90° γ≠90°	1(P)
5	Hexagonal	a= b≠ c	α=β= 90° γ=120°	1(P)
6	Monoclinic	a≠ b≠ c	α=β= 90° γ≠90°	2(P,C)
7	Triclinic	a≠ b≠ c	α≠β≠ γ ≠90°	1(P)

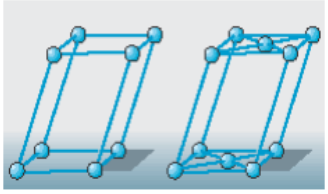
cubic



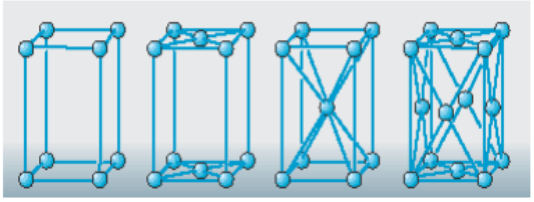
rhombohedral



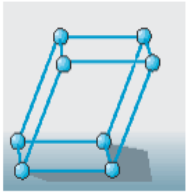
tetragonal



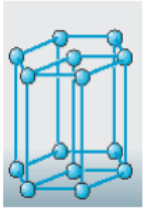
monoclinic



orthorhombic



triclinic



hexagonal

Activa
o to P

Unit Cell Charateristics

Effective number of atoms per unit cell (Z):

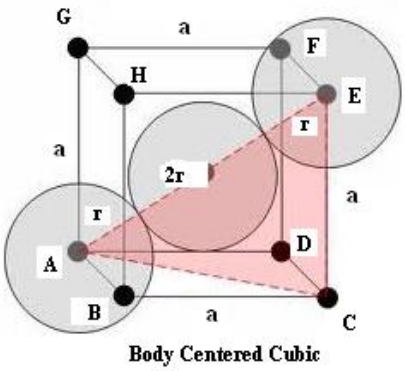
A unit cell is a part of an infinite scheme, and not an isolated entity. Therefore several adjacent cells share each lattice point. As a result the basis attached to a lattice site contributes only a fractionof its mass and volume to one unit cell. The effective number of atoms per unit cell is given by,

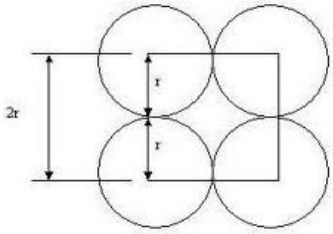
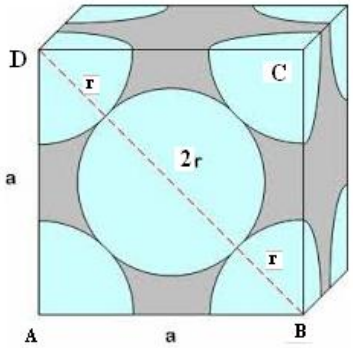
$$Z = Z_B + \frac{Z_F}{2} + \frac{Z_C}{8}$$

Where, Z_B **= Number of body centered atom**
Z_F **= Number of Face centered atoms**
Z_C **= Number of Corner atoms**

SC	BCC	FCC
For, SC $Z_B = 0\ , Z_F = 0 \& Z_C = 8$ $\therefore\ Z = Z_B + \frac{Z_F}{2} + \frac{Z_C}{8}$ $\therefore\ Z = 0 + \frac{0}{2} + \frac{8}{8}$ $\therefore\ Z = 1$	For, BCC $Z_B = 1\ , Z_F = 0 \& Z_C = 8$ $\therefore\ Z = Z_B + \frac{Z_F}{2} + \frac{Z_C}{8}$ $\therefore\ Z = 1 + \frac{0}{2} + \frac{8}{8}$ $\therefore\ Z = 2$	For, FCC $Z_B = 0\ , Z_F = 6 \& Z_C = 8$ $\therefore\ Z = Z_B + \frac{Z_F}{2} + \frac{Z_C}{8}$ $\therefore\ Z = 0 + \frac{6}{2} + \frac{8}{8}$ $\therefore\ Z = 4$

Atomic Radius:

SC	**BCC**	**FCC**
For SC unit cell atoms are in contact along the edges of the	For Bcc cell atoms are in contact along the body diagonal of the cube.  Body Centered Cubic From Fig ΔACE	For FCC cell atoms are in contact along the face diagonal of the cube.

 cube. a = 2r  From fig. if 'a' is the edge of a unit cell and 'r' is the radius of the atom. $\therefore a = 2 \times r$ $\therefore \boldsymbol{r = \frac{a}{2}}$	$(AE)^2 = (AC)^2 + (CE)^2$ $(4\ r)^2 = [(AB)^2 + (BC)^2] + (CE)^2$ From ΔABC $(4\ r)^2 = [(a)^2 + (a)^2] + (a)^2$ $(4\ r)^2 = 3(a)^2$ Taking square root $4\ r = \sqrt{3}\ a$ $\boldsymbol{r = \frac{\sqrt{3}\ a}{4}}$	 Face centered Cubic From Fig ΔABD $(BD)^2 = (AB)^2 + (AD)^2$ $(4\ r)^2 = (a)^2 + (a)^2$ $(4\ r)^2 = 2(a)^2$ Taking square root $4\ r = \sqrt{2}\ a$ $r = (\sqrt{2}\ a)/4$ $\boldsymbol{r = \frac{a}{2\sqrt{2}}}$

<u>Atomic packing Fraction:</u>

The fraction of a space occupied by atoms in a unit cell is known as atomic packing fraction. It is defined as the ratio of volume of effective number of atoms in a unit cell to the total volume of unit cell.

$$APF = \frac{\frac{No\ of\ atoms}{Unit\ cell} \times (Volume\ of\ each\ atom)}{Volume\ of\ unit\ cell}$$

$APF = \frac{Zv}{V}$ Where, Z= Effective number of atoms per unit cell

V= Volume of unit Cell

v = Volume of each atom

SC	BCC	FCC
$APF = \frac{\frac{No\ of\ atoms}{Unit\ cell} \times (Volume}{Volume\ of\ unit}$ $APF = \frac{Zv}{V}$	$APF = \frac{\frac{No\ of\ atoms}{Unit\ cell} \times (Volume}{Volume\ of\ unit}$ $APF = \frac{Zv}{V}$	$APF = \frac{\frac{No\ of\ atoms}{Unit\ cell} \times (Volume}{Volume\ of\ unit}$ $APF = \frac{Zv}{V}$

For SC, Z= 1 and r= a/2 ∴ $a = 2r$ $\therefore APF = \frac{(\frac{4}{3}\pi r^3)}{a^3}$ $\therefore APF = \frac{(\frac{4}{3}\pi r^3)}{8r^3}$ $\therefore APF = \frac{\pi}{6}$ $\therefore APF = 0.52$	For BCC, Z= 2 and r= (√ 3 a)/4 ∴ $a = \frac{4r}{\sqrt{3}}$ $\therefore APF = \frac{2(\frac{4}{3}\pi r^3)}{a^3}$ $\therefore APF = \frac{2(\frac{4}{3}\pi r^3)}{\frac{64}{3\sqrt{3}}r^3}$ $\therefore APF = \frac{\pi\sqrt{3}}{8} = 0.68$	For FCC, Z= 4 and r = a/ $2\sqrt{2}$ ∴ $a = 2\sqrt{2} \times r$ $\therefore APF = \frac{4(\frac{4}{3}\pi r^3)}{a^3}$ $\therefore APF = \frac{4(\frac{4}{3}\pi r^3)}{(2\sqrt{2} \times r)^3}$ $\therefore APF = \frac{\pi}{3\sqrt{2}}$ $\therefore APF = 0.74$

<u>Void Space:</u>

The void space in unit cell is the vacant space left unutilized in the unit cell. It is generally expressed in percentage.

Void Space = (1 - APF)×100

SC	BCC	FCC
<u>Void Space:</u> **Void Space = (1 - APF)×100** **Void Space = (1 – 0.52)×100** **Void Space = 48%**	**<u>Void Space:</u>** **Void Space = (1 - APF)×100** **Void Space = (1 – 0.68)×100** **Void Space =32%**	**<u>Void Space:</u>** **Void Space = (1 - APF)×100** **Void Space = (1 – 0.74)×100** **Void Space =26%**

Thus from the above it is observed that FCC structure possesses least percent void space and maximum packing density amongst all cubic structures.

Density (ρ):

$$\boldsymbol{Density} = \rho = \frac{\boldsymbol{Mass}}{\boldsymbol{Volume}} = \frac{\boldsymbol{ZW}}{\boldsymbol{V}}$$

Where, Z= No. of atoms per unit cell.

and W= mass of each atom.

$W=M/N_A$

Where M=molecular weight

N_A=Avogadro's number.

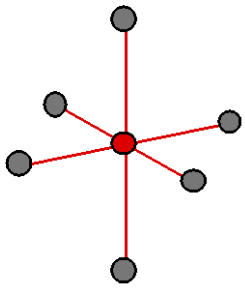
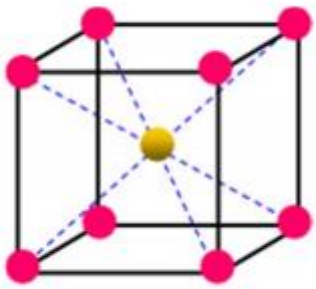
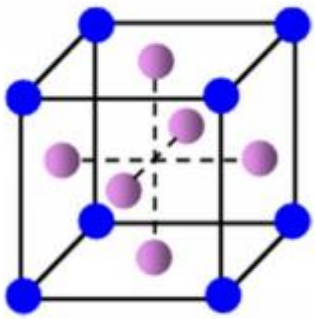
$$\rho=\frac{\boldsymbol{ZM}}{\boldsymbol{N_A.V}}$$

SC	BCC	FCC
As, $Density = \rho = \frac{ZM}{N_A.V}$ **For SC, Z=1** $\therefore \rho = \frac{1M}{N_A.V}$ $\therefore \rho = \frac{M}{N_A.V}$	As, $Density = \rho = \frac{ZM}{N_A.V}$ **For BCC, Z=2** $\therefore \rho = \frac{2M}{N_A.V}$ $\therefore \rho = \frac{2M}{N_A.V}$	As, $Density = \rho = \frac{ZM}{N_A.V}$ **For, FCC Z=4** $\therefore \rho = \frac{4M}{N_A.V}$ $\therefore \rho = \frac{4M}{N_A.V}$

Thus from the above equation it Show that atomic density of BCC is double than SC unit cell.

Coordination number

Ans: Coordination number of an atom in a crystal is the number of nearest neighboring atoms which are simultaneously in contact with that atom. It signifies the tightness of packing of atoms in the crystal.(.Greater the co-ordination no, the atoms are said to be closely packed.)

SC	BCC	FCC
• **Coordination No. = 6 (# nearest neighbors) for each atom as seen** 	Coordination number=8 The body centered atom is in contact with all the eight corner atoms. Each corner atom is shared by eight unit cells and hence, each of these atoms is in touch with eight body centered atoms. 	Coordination number=12 The face centered atom, in the front face is in contact with four corner atoms and four other face-centered atoms behind it (two sides, top and bottom) and is also touching four face-centered atoms of the unit cell in front of it 

Miller Indices(MI):

Miller indices are the reciprocals of intercepts made by the crystal planes on the crystallographic axes when reduced to smallest integers.

Important features of Miller indices:

• Miller indices represent a set of parallel equidistant planes.

• All the parallel equidistant planes have the same Miller indices.

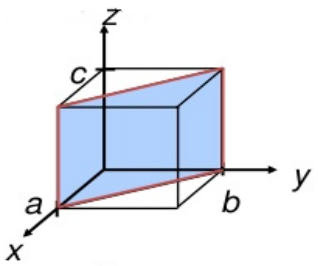
• If a plane is parallel to any axis, then the plane intersects that axis at infinity and Miller indices along that direction is zero.

• If the miller indices of the two planes have the same ratio (844,422,211), then the planes are parallel to each other.

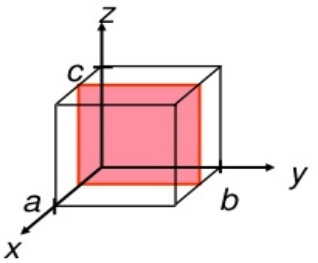
• If a plane cuts an axis on the –ve side of the origin, then the corresponding index is – ve, and is indicated by placing a minus sign above the index. Ex: if a plane cuts –ve y-axis, then the miller index of the plane is (h $\bar{k}$ l)of the translation vectors. (2) Take the reciprocal of the intercepts (1/l1, 1/l2 1/l3) and multiple by the smallest constant that makes them integers (h, k, l). Miller Indices can also be used to label the group of equivalent planes {h, k, l}. Examples: (2, 3, 3)

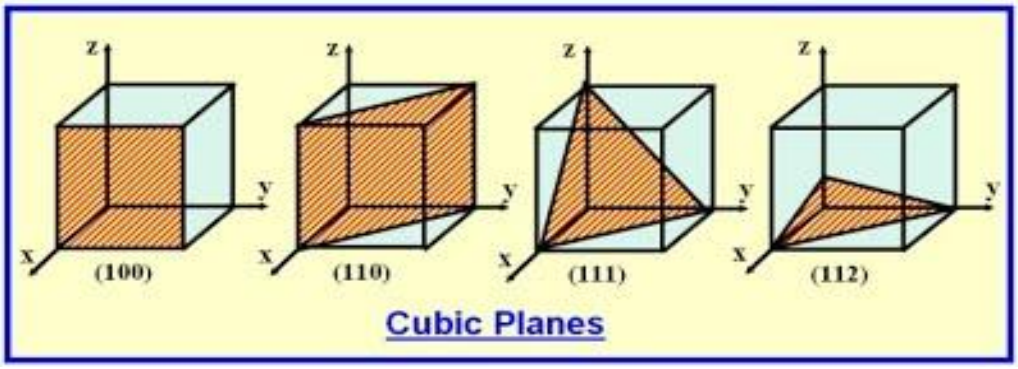
Q. :Find Miller indices of given planes

example		a	b	c
1.	Intercepts	1	1	∞
2.	Reciprocals	1/1	1/1	1/∞
		1	1	0
3.	Reduction (x LCM)	1	1	0
4.	Miller Indices	(110)		



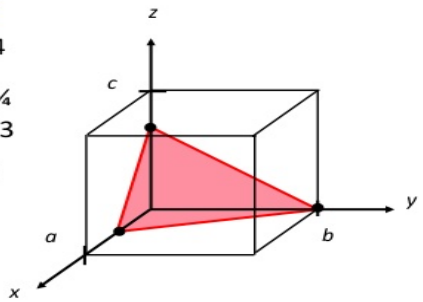
example		a	b	c
1.	Intercepts	1/2	∞	∞
2.	Reciprocals	1/½	1/∞	1/∞
		2	0	0
3.	Reduction x LCM	2	0	0
4.	Miller Indices	(100)		





example

		a	b	c
1.	Intercepts	1/2	1	3/4
2.	Reciprocals	1/½	1/1	1/¾
		2	1	4/3
3.	Reduction (X by LCM)	6	3	4
4.	Miller Indices	(634)		



Bragg's Law:

Statement: It states that, when X-rays are incident on crystalline planes at particular angel Θ, then the reflected rays from these planes interface constructively and gives maximum intensity along a particular direction at angle between plane and X-rays is called Bragg's angle or glancing angel.

The angle 2Θ is called diffraction angle. Bragg's equation is given as

$$2d\sin\Theta = n\lambda$$

Where, d-is interplaner distance.

n-is order of diffraction

λ-is wavelength of incident X-rays.

Proof:

Let us consider a series of Bragg's planes separated by distance (d).

Let beam of monochromatic X-rays be incident on the plans MN and PQ along AB and DE at an angle (Θ). The scattered (diffracted) beam undergoes along BCand EF.

The diffracted rays BC and EF interfere constructively if Θ=Θ' [(∠CBN)= Θ']

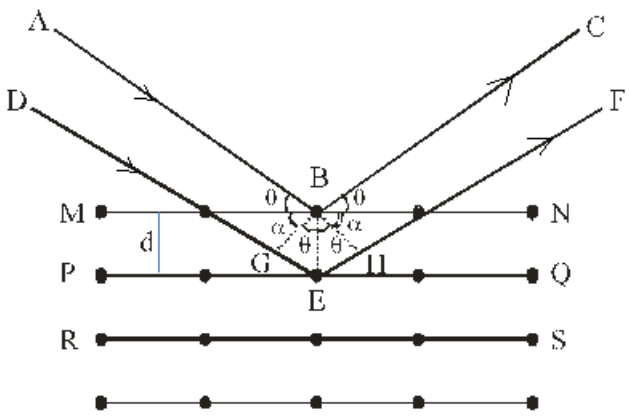
This condition is achieve only when path difference = nλ.

Path difference =Δ = nλ ---------- (1)

Path difference of rays BC and EF is

Δ=GE+EH from Eq(1) we get,

GE+EH= nλ ---------- (2)



Let us draw perpendicular from plane MN to plane PQ and from rays AB to ray DG, from ray BC to plan BA.

$\therefore BE=d.; \quad MBA=\Theta$

Let us assume that $\angle MBG=\alpha$.

$\angle ABG=\angle MBA+\angle MBG$

$90°=\Theta+\alpha$ ---------- (3)

$\angle MBE=\angle MBG+\angle GBE$

$90°=\alpha+\angle GBE$ ---------- (4)

From (3) and (4)$\rightarrow\angle GBE=\Theta$

Similarly, $\angle EBH=\Theta$

In Δ GBE, $\quad Sin\Theta=(GE/BE)=(GE/d)$

$$\left.\begin{array}{l} GE=dSin\Theta \\ \text{Similarly, } EH=dSin\Theta \end{array}\right\} \text{---------- (5)}$$

Substituting Eq. (5) in Eq. (2)

We get, $\quad dSin\Theta+dSin\Theta=n\lambda$

$$\boxed{2dSin\Theta=n\lambda} \text{ ---------- (6)}$$

Eq. (6) is known as Bragg's Law.

Application

1) Knowing the inter planer distance, structure of crystal can be known.
2) To find the wavelength of monochromatic radiation